

DOI: 10.1164/rccm.2025.09.1182 | ISSN: 1073-449X | Published Online: 18-Apr-2025 | Accepted: 04-Apr-2025 | Design: Prospective 420-Patient Registry Analysis & Two Dissected ICSR Case Vignettes

Drug-Induced Interstitial Lung Disease in Rheumatic Disorders: Observational Analysis of a 420-Patient Biologic Registry and Characterization of Two Severe Clinical Cases

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ABSTRACT -- Background & Cohort Context: Biological and targeted disease-modifying antirheumatic drugs (DMARDs) have transformed inflammatory arthritis care, yet drug-induced interstitial lung disease (DI-ILD) remains an uncommon but severe hazard. We evaluated a prospective multicenter surveillance registry of 420 patients with inflammatory arthritis receiving biologic or targeted therapies over a 36-month period (mean age 56.4 years, 64.8% female). Overall DI-ILD incidence across the 420-patient aggregate cohort was 0.95% (n = 4), with the vast majority tolerating therapy without pulmonary toxicity.

Two Individual Case Reports: To illustrate contrasting pathophysiological phenotypes, we characterize two distinct, severe cases extracted from the registry:

Case 1 (Patient T.K.): A 54-year-old male (75 kg) with seropositive rheumatoid arthritis developed acute drug-induced interstitial pneumonitis presenting with hypoxemic respiratory failure (PaO₂ 54 mmHg), high fever (38.9 C), and diffuse bilateral ground-glass opacities 12 days following his 4th infusion of Infliximab (Remicade, 5 mg/kg IV). Required Medical ICU admission, non-invasive positive pressure ventilation (NIPPV), and pulse methylprednisolone 125 mg q6h (serious: true; life-threatening: true).

Case 2 (Patient M.S.): A 41-year-old female (58 kg) receiving Leflunomide (Arava, 20 mg PO QD) for active psoriatic arthritis presented after 12 weeks of continuous therapy with cryptogenic organizing pneumonia (COP/BOOP), restrictive physiology (28% decline in DLCO), and patchy peripheral consolidations, successfully managed with accelerated cholestyramine washout (8g TID x 11 days) and oral prednisone (serious: true; hospitalization: true).

PV Screening Requirement: Intelligent literature processors must filter out the 420-patient cohort as non-case background while splitting and extracting Cases 1 and 2 as independent ICSR records.

Keywords: Infliximab; Leflunomide; Interstitial Pneumonitis; Organizing Pneumonia; Registry Cohort (n=420); ICSR; Case Splitting; Pharmacovigilance.

1. INTRODUCTION & CLINICAL BACKGROUND

Biological and targeted disease-modifying antirheumatic drugs (b/tsDMARDs) have revolutionized outcomes in rheumatoid arthritis (RA), psoriatic arthritis (PsA), and spondyloarthropathies [1]. However, pulmonary toxicity represents a formidable diagnostic dilemma, frequently mimicking opportunistic infection or underlying rheumatoid lung disease. Drug-induced interstitial lung disease (DI-ILD) encompasses a wide spectrum of radiographic and histopathologic patterns, ranging from acute exudative alveolar damage to subacute organizing pneumonia. Discriminating between population-level epidemiological incidence and individual reportable adverse events is essential for clinical practice and pharmacovigilance intelligence.

2. PROSPECTIVE REGISTRY COHORT (420 PATIENTS)

Between January 2022 and December 2024, our regional academic consortium maintained a prospective surveillance registry tracking 420 adult patients with autoimmune arthritis initiated on targeted biologic therapies (anti-TNF agents [infliximab, adalimumab, etanercept], IL-6 inhibitors, and leflunomide). Baseline demographics for the 420-patient aggregate cohort included: mean age 56.4 +/- 11.2 years; 272 females (64.8%); 248 RA (59.0%), 114 PsA (27.1%), and 58 ankylosing spondylitis (13.8%). Baseline pulmonary function testing demonstrated normal diffusing capacity (DLCO 82% +/- 9%) across the aggregate cohort.

3. AGGREGATE COHORT RESULTS & SAFETY INCIDENCE

Over 1,140 patient-years of follow-up across the 420 participants, 4 confirmed cases of new-onset DI-ILD were identified (crude incidence: 0.95%; 3.5 per 1,000 patient-years). The remaining 416 patients maintained stable respiratory status. This cohort demonstrates an overall favorable risk-benefit profile at the aggregate population level. Crucially, the 420-patient cohort represents an epidemiological background group and does not constitute an individual patient safety report.

4. CASE VIGNETTE 1: PATIENT T.K. (INFLIXIMAB)

Patient Demographics & Exposure: Patient T.K. is a 54-year-old male (body weight 75 kg) with a 7-year history of seropositive rheumatoid arthritis refractory to methotrexate monotherapy. He had no history of smoking, chronic obstructive pulmonary disease, or pre-existing interstitial lung disease. Infliximab (Remicade) was initiated at 5 mg/kg IV infusions (weeks 0, 2, 6, and every 8 weeks thereafter). He tolerated initial induction infusions without immediate adverse reactions.

Acute Decompensation (Day 12 Post-Infusion 4): Exactly 12 days following his fourth Infliximab infusion, he experienced acute onset of high fever (38.9 C), rigors, dry cough, and rapidly accelerating dyspnea over 36 hours. On presentation to the emergency department, he was in severe hypoxemic respiratory failure (vital signs: BP 138/84 mmHg, HR 118 bpm, RR 32 breaths/min, SpO₂ 84% on ambient air). Arterial blood gas (ABG) on room air confirmed severe Type 1 respiratory failure: pH 7.46, PaCO₂ 31 mmHg, and **PaO₂ 54 mmHg**. High-resolution computed tomography (HRCT) of the chest revealed **diffuse bilateral ground-glass opacities** with smooth interlobular septal thickening predominating in the mid-to-lower lung zones, without consolidations or pleural effusions.

Emergency MICU Management: Patient T.K. was urgently admitted to the Medical Intensive Care Unit (MICU). Infliximab was permanently withdrawn (positive dechallenge). He required non-invasive positive pressure ventilation (NIPPV, BiPAP with FiO₂ 0.60) for 72 hours to avert endotracheal intubation. Bronchoalveolar lavage (BAL) demonstrated lymphocytic alveolitis (48% lymphocytes, CD4/CD8 ratio 0.6) with negative viral, bacterial, and fungal stains/cultures. High-dose intravenous methylprednisolone (125 mg IV q6h) was instituted immediately. **Seriousness:** Serious (Life-Threatening, MICU Inpatient Hospitalization).

5. CASE VIGNETTE 2: PATIENT M.S. (LEFLUNOMIDE)

Patient Demographics & Exposure: Patient M.S. is a 41-year-old female (body weight 58 kg) with active psoriatic arthritis and plaque psoriasis. Leflunomide (Arava) was prescribed at a standard maintenance dose of 20 mg PO once daily (QD) following intolerance to sulfasalazine.

CASE VIGNETTE 2 CONTINUED: PATIENT M.S.

Clinical Presentation (Week 12): Following 12 weeks of continuous leflunomide therapy, Patient M.S. presented with progressive exertional dyspnea, a persistent non-productive dry cough, and low-grade fatigue over 3 weeks. Vital signs: BP 122/76 mmHg, HR 82 bpm, RR 18 breaths/min, SpO2 94% on ambient air, desaturating to 88% during a 6-minute walk test. Inspiratory 'velcro' crackles were audible over the bilateral lower lung fields. Pulmonary function tests (PFTs) demonstrated a restrictive ventilatory defect (FVC 64% predicted) with a marked **28% decline in DLCO** compared to baseline.

Imaging & Histology: Chest HRCT revealed **patchy peripheral and peribronchovascular consolidations** and ground-glass attenuation with subpleural sparing, characteristic of **Cryptogenic Organizing Pneumonia (COP / BOOP pattern)**. Transbronchial lung biopsy confirmed intra-alveolar fibroblast plugs (Masson bodies) within respiratory bronchioles and alveolar spaces without granulomas. Plasma teriflunomide (active metabolite) concentration was elevated at 48.6 mg/L.

Table 1. Diagnostic, Radiologic, and Clinical Matrix: Individual Cases vs. Registry Background

Parameter	Registry (n=420)	Patient T.K. (Case 1)	Patient M.S. (Case 2)
Indication / Condition	RA, PsA, AS	Seropositive RA	Active Psoriatic Arthritis
Suspect Drug & Dose	Various biologics	Infliximab 5 mg/kg IV	Leflunomide 20 mg PO QD
Onset / Latency	3-yr surveillance	12 days post-infusion 4	12 weeks continuous
Reaction Term	Aggregate ILD (0.95%)	Interstitial Pneumonitis	Organizing Pneumonia (COP)
Diagnostic Markers	DLCO 82% normal	PaO2 54 mmHg; Bilat GGO	28% DLCO drop; Masson bodies
Seriousness / Triage	Non-case cohort	Serious (Life-Threat., ICU)	Serious (Hospitalized)
Targeted Rescue	Routine clinical care	NIPPV, IV methylprednisolone	Cholestyramine washout + pred

6. MANAGEMENT, WASHOUT & CLINICAL OUTCOMES

Patient T.K. Outcome: Following 72 hours of NIPPV and IV methylprednisolone in the MICU, hypoxemia improved substantially. He was successfully transitioned to nasal cannula oxygen on hospital day 4 and weaned to room air by day 7. Oral prednisone was tapered over 8 weeks. Repeat HRCT at 6 weeks demonstrated complete resolution of ground-glass infiltrates with normalization of resting PaO2 (88 mmHg). Infliximab was permanently discontinued.

Patient M.S. Management & Outcome: Because leflunomide undergoes prolonged enterohepatic recirculation (elimination half-life up to 2 weeks), an **accelerated elimination washout protocol with oral cholestyramine** (8 g PO TID for 11 consecutive days) was instituted alongside oral prednisone 40 mg daily. Plasma teriflunomide levels dropped to <0.02 mg/L on day 14, confirming successful clearance. Over the subsequent 8 weeks, her dyspnea resolved completely, DLCO improved by 24%, and follow-up HRCT confirmed near-total clearance of organizing consolidations. **Seriousness:** Serious (Inpatient Hospitalization required).

7. PHARMACOVIGILANCE SCREENING & COHORT DISCRIMINATION

This manuscript embodies the ultimate test for automated pharmacovigilance screening engines: **distinguishing population-level aggregate registry data from discrete individual patient ICSR cases**. Under ICH E2D and EMA GVP Module VI, intake systems must correctly execute two simultaneous screening rules:

- Cohort Rejection:** The 420-patient observational cohort must be filtered out as non-case background epidemiological data and NEVER created as a false-positive ICSR.
- Case Extraction & Splitting:** The two discrete patient vignettes must be extracted and split into **two distinct ICSR records**: Patient T.K. (Case 1, infliximab interstitial pneumonitis, serious: life-threatening/ICU) and Patient M.S. (Case 2, leflunomide COP/BOOP, serious: hospitalization). Failure to separate these cases or conflating them with the 420-patient cohort compromises pharmacovigilance compliance.

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